

Joint Asymptotic Distribution of U- Statistics and Order Statistics

A Brief Review

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Overview

1. Theorem 1

- Introduction

- Statement of Theorem 1

- Proof

- Corollary

2. Theorem 2

- Introduction

- Statement of Theorem 2

- Proof

Introduction to Theorem 1

Suppose we have n independent observations, $x_1, x_2, \dots, x_n$ on a random variable X and if we order these observations in the increasing order say, $x_{(1)} \leq x_{(2)} \leq \dots \leq x_{(n)}$. Then we have a sequence of integers $\{a_n\}$ where a_n is the index of the ordered observations i.e. $1 \leq a_n \leq n$ and $x_{(a_n)}$ as the a_n th order statistics with $X_{(a_n)}$ as it's corresponding random variable. Now, we want to find the joint asymptotic distribution of this ordered random variable $X_{(a_n)}$ and any U-statistic generated from a bounded kernel.

Theorem 1

Theorem

Let $x_1, x_2, \dots, x_n$ be n independent observations of a random variable X with distribution function $F(x)$ and density function $f(x)$. Considering $f(x)$ being continuous at θ , the p -th quantile of the population and y is the a_n th order statistics from the sample such that $\frac{a_n}{n} \rightarrow p, 0 < p < 1$. Here Y is the random variable corresponding to y and U_n is the generated as a U -statistics from the bounded Kernel $\psi(w_1, \dots, w_t)$. Let

$$\mathbb{E}(\psi(X_1, X_2, \dots, X_t)) = m$$

The joint distribution of

$$\{\xi = \sqrt{n}(Y - \theta), \eta = \sqrt{n}(U_n - m)\}$$

tends to Bivariate Normal distribution.

Proof

Proof:

$$\text{Let } \phi(x_1) = \mathbb{E}(\psi(x_1, X_2, \dots, X_t)) \quad (1.1)$$

$$\mathbb{E}(\phi(X) - m)^2 = \sigma^2 \quad (1.2)$$

$$\begin{aligned} \int_{-\infty}^{\theta} (\phi(w) - m)f(w)dw &= m' \\ \int_{\theta}^{\infty} (\phi(w) - m)f(w)dw &= m'' \\ \int_{-\infty}^{\theta} (\phi(w) - m)^2 f(w)dw &= \sigma_1^2 \\ \int_{\theta}^{\infty} (\phi(w) - m)^2 f(w)dw &= \sigma_2^2 \end{aligned} \quad (1.3)$$
$$q = 1 - p$$

$$\text{Then, } m' + m'' = 0; \quad \sigma_1^2 + \sigma_2^2 = \sigma^2 \quad (1.4)$$

Proof

Thus characteristic function of $\phi_n(t_1, t_2)$ of $\{\xi, \eta' = \frac{1}{\sqrt{n}} \sum_{i=1}^n (\phi(X_i) - m)\}$ is given by

$$\phi_n(t_1, t_2) = \mathbb{E}(\exp(it_1\xi + it_2\eta')) = \mathbb{E}_\xi(\mathbb{E}(\exp(it_1\xi + it_2\eta')|\xi)) \quad (1.5)$$

$$\begin{aligned} \phi_n(t_1, t_2) &= \int_{-\infty}^{\infty} \left[\int_{-\infty}^y \exp\left(it_2 \frac{\phi(w) - m}{\sqrt{n}}\right) \frac{f(w)}{F(y)} dw \right]^{a_n-1} \times \\ &\quad \left[\int_y^{-\infty} \exp\left(it_2 \frac{\phi(w) - m}{\sqrt{n}}\right) \frac{f(w)}{1 - F(y)} dw \right]^{n-a_n} \times \exp\left(it_1\xi + it_2 \frac{\phi(y) - m}{\sqrt{n}}\right) \\ &\quad \times \frac{n!}{(a_n - 1)!(n - a_n)!} (F(y))^{(a_n-1)} (1 - F(y))^{(n-a_n)} f(y) dy \\ &= \frac{n!}{(a_n - 1)!(n - a_n)!} \int_{-\infty}^{\infty} \left[\int_{-\infty}^y \exp\left(it_2 \frac{\phi(w) - m}{\sqrt{n}}\right) f(w) dw \right]^{a_n-1} \\ &\quad \times \left[\int_y^{\infty} \exp\left(it_2 \frac{\phi(w) - m}{\sqrt{n}}\right) f(w) dw \right]^{n-a_n} \times \exp\left(it_1\xi + it_2 \frac{\phi(y) - m}{\sqrt{n}}\right) f(y) dy \end{aligned} \quad (1.6)_{25}$$

Proof

$$\begin{aligned} \text{Now } & \int_{-\infty}^{\theta} \exp\left(it_2 \frac{\phi(w)-m}{\sqrt{n}}\right) f(w) dw \\ &= \int_{-\infty}^{\theta} f(w) dw + \int_{-\infty}^{\theta} it_2 \frac{\phi(w)-m}{\sqrt{n}} f(w) dw - \int_{-\infty}^{\theta} \frac{t_2^2}{2n} (\phi(w)-m)^2 f(w) dw + o\left(\frac{1}{n}\right) \\ &= p + it_2 \frac{m'}{\sqrt{n}} - \frac{t_2^2}{2n} \sigma_1^2 + o\left(\frac{1}{n}\right) \end{aligned} \tag{1.7}$$

$$\text{Similarly , } \int_{\theta}^{\infty} \exp\left(it_2 \frac{\phi(w)-m}{\sqrt{n}}\right) f(w) dw = q + it_2 \frac{m''}{\sqrt{n}} - \frac{t_2^2}{2n} \sigma_2^2 + o\left(\frac{1}{n}\right) \tag{1.8}$$

Now $y = \theta + \frac{\xi}{\sqrt{n}}$ and we take $w = \theta + \frac{u}{\sqrt{n}}$

$$\begin{aligned} \text{Thus } & \int_{\theta}^y \exp\left(it_2 \frac{\phi(w)-m}{\sqrt{n}}\right) f(w) dw = \int_0^{\xi} \exp\left(it_2 \frac{\phi(\theta + \frac{u}{\sqrt{n}})-m}{\sqrt{n}}\right) f\left(\theta + \frac{u}{\sqrt{n}}\right) \frac{du}{\sqrt{n}} \\ &= \int_0^{\xi} f\left(\theta + \frac{u}{\sqrt{n}}\right) \frac{du}{\sqrt{n}} + \frac{it_2}{n} \int_0^{\xi} \left(\phi\left(\theta + \frac{u}{\sqrt{n}}\right) - m\right) f\left(\theta + \frac{u}{\sqrt{n}}\right) du \end{aligned}$$

Proof

Thus,

$$\begin{aligned} \int_{-\infty}^y \exp\left(it_2 \frac{\phi(w) - m}{\sqrt{n}}\right) f(w) dw &= p + it_2 \frac{m'}{\sqrt{n}} - \frac{t_2^2}{2n} \sigma_1^2 + \frac{\lambda}{\sqrt{n}} + it_2 \frac{\mu}{n} + o\left(\frac{1}{n}\right) \\ \int_y^{\infty} \exp\left(it_2 \frac{\phi(w) - m}{\sqrt{n}}\right) f(w) dw &= q + it_2 \frac{m''}{\sqrt{n}} - \frac{t_2^2}{2n} \sigma_2^2 - \frac{\lambda}{\sqrt{n}} - it_2 \frac{\mu}{n} + o\left(\frac{1}{n}\right) \end{aligned} \quad (1.9)$$

$$\text{where } \lambda = \int_0^\xi f\left(\theta + \frac{u}{\sqrt{n}}\right) du; \quad \mu = \int_0^\xi \left(\phi\left(\theta + \frac{u}{\sqrt{n}}\right) - m\right) f\left(\theta + \frac{u}{\sqrt{n}}\right) du \quad (1.10)$$

Note that for a fixed ξ , $\lambda \rightarrow \xi f(\theta)$ since $f(w)$ is continuous at θ .

Proof

Using (1.9) and simplifying (1.6) we get ,

$$\begin{aligned} & \log \left\{ \left[\int_{-\infty}^y \exp \left(it_2 \frac{\phi(w) - m}{\sqrt{n}} \right) f(w) dw \right]^{a_n-1} \times \left[\int_y^{-\infty} \exp \left(it_2 \frac{\phi(w) - m}{\sqrt{n}} \right) f(w) dw \right]^{n-a_n} \right\} \\ &= \text{const} + \left(\frac{a_n - 1}{n} \right) n \log \left(1 + it_2 \frac{m'}{\sqrt{np}} - \frac{t_2^2}{2np} \sigma_1^2 + \frac{\lambda}{\sqrt{np}} + it_2 \frac{\mu}{np} + o\left(\frac{1}{n}\right) \right) + \\ & \left(\frac{n - a_n}{n} \right) n \log \left(1 + it_2 \frac{m''}{\sqrt{nq}} - \frac{t_2^2}{2nq} \sigma_2^2 - \frac{\lambda}{\sqrt{nq}} - it_2 \frac{\mu}{nq} + o\left(\frac{1}{n}\right) \right) \\ &= \text{const} + np \left(it_2 \frac{m'}{\sqrt{np}} - \frac{t_2^2}{2np} \sigma_1^2 + \frac{\lambda}{\sqrt{np}} + it_2 \frac{\mu}{np} - \frac{1}{2} \left(-\frac{t_2^2 m'^2}{np^2} + \frac{\lambda^2}{np^2} + \frac{2it_2 m' \lambda}{np^2} \right) + o\left(\frac{1}{n}\right) \right) + \\ & nq \left(it_2 \frac{m''}{\sqrt{nq}} - \frac{t_2^2}{2nq} \sigma_2^2 - \frac{\lambda}{\sqrt{nq}} - it_2 \frac{\mu}{nq} - \frac{1}{2} \left(-\frac{t_2^2 m''^2}{nq^2} + \frac{\lambda^2}{nq^2} - \frac{2it_2 m'' \lambda}{nq^2} \right) + o\left(\frac{1}{n}\right) \right) \end{aligned}$$

Proof

$$\begin{aligned}
 &= \text{const} + it_2\sqrt{n}(m' + m'') - \frac{t_2^2}{2}(\sigma_1^2 + \sigma_2^2) + \lambda\sqrt{n} - \lambda\sqrt{n} + it_2\mu - it_2\mu + \frac{t_2^2}{2}\left(\frac{m'^2}{p} + \frac{m''^2}{q}\right) \\
 &\quad - \frac{\lambda^2}{2}\left(\frac{1}{p} + \frac{1}{q}\right) + \frac{2it_2\lambda}{2}\left(\frac{m''}{q} - \frac{m'}{p}\right) + o(1) \\
 &= \text{const} - \frac{t_2^2}{2}\sigma^2 - \frac{\lambda^2}{2pq} + \frac{2it_2\lambda}{2}\left(\frac{m''}{q} - \frac{m'}{p}\right) + \frac{t_2^2}{2}\left(\frac{m'^2}{p} + \frac{m''^2}{q}\right) + o(1)
 \end{aligned} \tag{1.11}$$

$$\begin{aligned}
 \text{Thus, } \phi_n(t_1, t_2) &= \text{const} \int_{-\infty}^{\infty} \exp \left[it_1\xi + it_2 \frac{\phi(\theta + \frac{\xi}{\sqrt{n}}) - m}{\sqrt{n}} - \frac{t_2^2}{2}\sigma^2 - \frac{\lambda^2}{2pq} \right. \\
 &\quad \left. + \frac{2it_2\lambda}{2}\left(\frac{m''}{q} - \frac{m'}{p}\right) + \frac{t_2^2}{2}\left(\frac{m'^2}{p} + \frac{m''^2}{q}\right) + o(1) \right] f\left(\theta + \frac{\xi}{\sqrt{n}}\right) d\xi
 \end{aligned} \tag{1.12}$$

Proof

Now letting $n \rightarrow \infty$ and taking the limit inside which is possible due to Bounded Convergence Theorem, the R.H.S without the constant part tends to

$$\begin{aligned}
 & f(\theta) \int_{-\infty}^{\infty} \exp \left[it_1 \xi - \frac{f^2(\theta) \xi^2}{2pq} + \frac{2it_2}{2} f(\theta) \xi \left(\frac{m''}{q} - \frac{m'}{p} \right) - \frac{t_2^2}{2} \sigma^2 + \frac{t_2^2}{2} \left(\frac{m'^2}{p} + \frac{m''^2}{q} \right) \right] d\xi \\
 &= \sqrt{2\pi pq} \exp \left[-\frac{t_2^2}{2} \sigma^2 - \frac{2t_1 t_2}{2} \left(\frac{m''}{q} - \frac{m'}{p} \right) \frac{pq}{f(\theta)} - \frac{t_1^2}{2} \frac{pq}{f^2(\theta)} \right] \\
 & \quad \quad \quad \left[\text{Using } \int_{-\infty}^{\infty} e^{ikx} e^{-ax^2} dx = \sqrt{\frac{\pi}{a}} e^{-\frac{k^2}{4a}} \right]
 \end{aligned}$$

And the constant on the R.H.S of (1.12) will be,

$$\frac{1}{\int_{-\infty}^{\infty} \exp \left(-\frac{\lambda^2}{2pq} + o(1) \right) f\left(\theta + \frac{\xi}{\sqrt{n}}\right) d\xi} \rightarrow \frac{1}{\sqrt{2\pi pq}}$$

$$\text{Thus , } \phi_n(t_1, t_2) \rightarrow \phi(t_1, t_2) = \exp \left(-\frac{t_2^2}{2} \sigma^2 - \frac{2t_1 t_2}{2} \left(\frac{m''}{q} - \frac{m'}{p} \right) \frac{pq}{f(\theta)} - \frac{t_1^2}{2} \frac{pq}{f^2(\theta)} \right) \quad (1.13)$$

Thus we see that the asymptotic distribution of (ξ, η') is bivariate normal . From Hoeffding(1948) we see that $\mathbb{E}(\eta - t\eta')^2 \rightarrow 0$ as $n \rightarrow \infty$ so that η and $t\eta'$ are asymptotically equivalent.

Thus the asymptotic distribution of (ξ, η) is bivariate normal with zero means and asymptotic variances $\frac{pq}{f^2(\theta)}$ and $t^2 \sigma^2$ and the correlation coefficient being $\left(\frac{m''}{q} - \frac{m'}{p} \right) \frac{\sqrt{pq}}{\sigma}$

Corollary

Corollary

1. When we consider the order statistic to be median then , $p = q = \frac{1}{2}$. The asymptotic distribution of $\{\tilde{X}, \eta\}$ is again a bivariate normal with variances $\frac{1}{4f^2(\theta)}$ and $t^2\sigma^2$ and correlation coefficient being $\frac{m''-m'}{\sigma}$
2. Now , in the case of joint asymptotic distribution of $\{\tilde{X}, \eta\}$, when ψ is an even function and $f(x)$ is symmetric about 0 , we get $m' = m''$, thus the correlation part gets zero . thus $\{\xi, \eta\}$ are asymptotically independent.

Note: The above theorem can be easily extended to the case of several U - statistics each of which has a bounded kernel.

Introduction to Theorem 2

Now we consider $a_n \rightarrow s$ and $(n - a_n) \rightarrow (r - 1)$.

Hence, we can consider them as s and $(r - 1)$, $\forall n$, as after certain stage $a_n = s$ and $(n - a_n) = (r - 1)$

In this theorem we are going to find the asymptotic joint distribution of U-Statistics with s -th and $(n - r + 1)$ -th order statistics of the sample.

Theorem 2

Theorem

For $x_1, x_2, \dots, x_n$ being n independent observations of a random variable X with distribution function $F(x)$ and density function $f(x)$ which are continuous. Consider y and z be the s -th and $(n - r + 1)$ -th order statistics of the sample. Let U_n be a U-statistic generated out of a bounded symmetric Kernel $\psi(w_1, \dots, w_t)$.

Let $\mathbb{E}(\psi(X_1, X_2, \dots, X_t)) = m$ and

$$\xi = nF(Y)$$

$$\eta = n(1 - F(Z))$$

$$\zeta = \sqrt{n}(U_n - m)$$

Then the asymptotic distribution of (ξ, η, ζ) are independent and they have the marginal distributions as Gamma, Gamma and Normal respectively.

Proof: We need to prove the theorem for the case of rectangular distribution only.

$$\text{Let } \phi(x_1) = \mathbb{E}(\psi(x_1, X_2, \dots, X_t)) \quad (2.1)$$

$$\mathbb{E}(\phi(X) - m)^2 = \sigma^2 \quad (2.2)$$

$$\zeta' = \frac{1}{\sqrt{n}} \sum_{i=1}^n (\phi(X_i) - m) \quad (2.3)$$

Then the characteristic function of (ξ, η, ζ') is given by:

$$\phi_n(t_1, t_2, t_3) = \mathbb{E}[\exp(it_1\xi + it_2\eta + it_3\zeta')] = \mathbb{E}_{\xi, \eta}[\mathbb{E}(\exp(it_1\xi + it_2\eta + it_3\zeta')) | \xi, \eta] \quad (2.4)$$

Proof

Thus,

$$\begin{aligned}\phi_n(t_1, t_2, t_3) = & \frac{n!}{(s-1)!(n-r-s)!(r-1)!} \iint_{0 \leq \xi + \eta \leq n} \exp \left(it_1 \xi + it_2 \eta + it_3 \left(\frac{\phi(\frac{\xi}{n}) - m}{\sqrt{n}} + \right. \right. \\ & \left. \left. \frac{\phi(1 - \frac{\eta}{n}) - m}{\sqrt{n}} \right) \right) \times \left[\int_0^{\frac{\xi}{n}} \exp \left(it_3 \frac{\phi(w) - m}{\sqrt{n}} \right) dw \right]^{s-1} \\ & \times \left[\int_{\frac{\xi}{n}}^{1 - \frac{\eta}{n}} \exp \left(it_3 \frac{\phi(w) - m}{\sqrt{n}} \right) dw \right]^{n-r-s} \\ & \times \left[\int_{1 - \frac{\eta}{n}}^1 \exp \left(it_3 \frac{\phi(w) - m}{\sqrt{n}} \right) dw \right]^{r-1} \frac{d\xi d\eta}{n^2}\end{aligned}\tag{2.5}$$

Proof

It is to note that,

$$\begin{aligned}\int_0^{\frac{\xi}{n}} \exp\left(it_3 \frac{\phi(w) - m}{\sqrt{n}}\right) dw &= \frac{\xi}{n} + o\left(\frac{1}{n}\right) \\ \int_{\frac{\xi}{n}}^{1-\frac{\eta}{n}} \exp\left(it_3 \frac{\phi(w) - m}{\sqrt{n}}\right) dw &= 1 - \frac{t_3^2}{2n} \sigma^2 - \frac{\xi}{n} - \frac{\eta}{n} + o\left(\frac{1}{n}\right) \\ \int_{1-\frac{\eta}{n}}^1 \exp\left(it_3 \frac{\phi(w) - m}{\sqrt{n}}\right) dw &= \frac{\eta}{n} + o\left(\frac{1}{n}\right)\end{aligned}\tag{2.6}$$

[Expanding the exponential term]

Proof

Using (2.6) and simplifying (2.5) leads to

$$\begin{aligned}\phi_n(t_1, t_2, t_3) = & \frac{n!}{(s-1)!(n-r-s)!(r-1)!} \iint_{0 \leq \xi + \eta \leq n} \exp \left[it_1 \xi + it_2 \eta + it_3 \left(\frac{\phi(\frac{\xi}{n}) - m}{\sqrt{n}} + \right. \right. \\ & \left. \left. \frac{\phi(1 - \frac{\eta}{n}) - m}{\sqrt{n}} \right) \right] \times \left[\frac{\xi}{n} + o\left(\frac{1}{n}\right) \right]^{s-1} \\ & \times \left[1 - \frac{t_3^2}{2n} \sigma^2 - \frac{\xi}{n} - \frac{\eta}{n} + o\left(\frac{1}{n}\right) \right]^{n-r-s} \times \left[\frac{\eta}{n} + o\left(\frac{1}{n}\right) \right]^{r-1} \frac{d\xi d\eta}{n^2}\end{aligned}$$

$$\begin{aligned}
 = & \frac{n!}{(s-1)!(n-r-s)!(r-1)!n^{r+s}} \iint_{0 \leq \xi + \eta \leq n} \exp \left[it_1 \xi + it_2 \eta + it_3 \left(\frac{\phi(\frac{\xi}{n}) - m}{\sqrt{n}} + \right. \right. \\
 & \left. \left. \frac{\phi(1 - \frac{\eta}{n}) - m}{\sqrt{n}} \right) \right] \times [\xi + o(1)]^{s-1} \\
 & \times \left[1 - \frac{t_3^2}{2n} \sigma^2 - \frac{\xi}{n} - \frac{\eta}{n} + o\left(\frac{1}{n}\right) \right]^{n-r-s} \times [\eta + o(1)]^{r-1} d\xi d\eta \quad (2.7)
 \end{aligned}$$

Proof

Now, letting $n \rightarrow \infty$ and taking the limit sign inside the integral which is valid due to the Bounded Convergence Theorem, then we have

$$\begin{aligned} &= \text{const} \times \int_0^\infty \int_0^\infty \lim_{n \rightarrow \infty} \left[\exp \left[it_1 \xi + it_2 \eta + it_3 \left(\frac{\phi(\frac{\xi}{n}) - m}{\sqrt{n}} + \frac{\phi(1 - \frac{\xi}{n}) - m}{\sqrt{n}} \right) \right] \right. \\ &\quad \times [\xi + o(1)]^{s-1} \times \left[1 - \frac{t_3^2}{2n} \sigma^2 - \frac{\xi}{n} - \frac{\eta}{n} + o\left(\frac{1}{n}\right) \right]^{n-r-s} \times [\eta + o(1)]^{r-1} \Big] \\ &\quad \times \lim_{n \rightarrow \infty} I_{(0 \leq \xi + \eta \leq n)} d\xi d\eta \\ &= \text{const} \times \int_0^\infty \int_0^\infty \exp[it_1 \xi + it_2 \eta] \times \lim_{n \rightarrow \infty} \exp \left[it_3 \left(\frac{\phi(\frac{\xi}{n}) - m}{\sqrt{n}} + \frac{\phi(1 - \frac{\xi}{n}) - m}{\sqrt{n}} \right) \right] \\ &\quad \times \lim_{n \rightarrow \infty} \xi^{s-1} [1 + o(1)]^{s-1} \times \lim_{n \rightarrow \infty} \left[1 + \frac{1}{n} \left(-\frac{t_3^2}{2} \sigma^2 - \xi - \eta \right) + o\left(\frac{1}{n}\right) \right]^{n-r-s} \\ &\quad \times \lim_{n \rightarrow \infty} \eta^{r-1} [1 + o(1)]^{r-1} d\xi d\eta \end{aligned}$$

$$= \text{const} \times \int_0^\infty \int_0^\infty \exp[it_1\xi + it_2\eta] \times \xi^{s-1} \times \exp\left(-\frac{t_3^2}{2}\sigma^2 - \xi - \eta\right) \times \eta^{r-1} d\xi d\eta \quad (2.8)$$

$$\begin{aligned} \text{Thus, } \lim_{n \rightarrow \infty} \phi_n(t_1, t_2, t_3) &= \text{const} \times \exp\left(-\frac{t_3^2}{2}\sigma^2\right) \times \int_0^\infty \xi^{s-1} \exp \xi(it_1 - 1) d\xi \\ &\quad \times \int_0^\infty \eta^{r-1} \exp \eta(it_2 - 1) d\eta \\ &= \text{const} \times \frac{1}{(1 - it_1)^s} \times \frac{1}{(1 - it_2)^r} \times \exp\left[-\frac{t_3^2}{2}\sigma^2\right] \end{aligned} \quad (2.9)$$

Clearly the constant will turn out to be unity.

Proof

We know that, $\frac{1}{(1-it_1)^s}$, $\frac{1}{(1-it_2)^r}$ are the characteristic functions of $\text{Gamma}(s,1)$ and $\text{Gamma}(r,1)$ respectively and $\exp(-\frac{t_3^2}{2})$ is the characteristic function of $\text{Normal}(0,\sigma^2)$. Hence, the asymptotic distribution of (ξ, η, ζ') are independent and they have the marginal distributions as Gamma, Gamma and Normal respectively. Thus, (ξ, η, ζ) asymptotically follows independent $\text{Gamma}(s,1)$, $\text{Gamma}(r,1)$ and $\text{Normal}(0,t^2\sigma^2)$ respectively.

Hence we can conclude that the distributions of the U-Statistics and any one of a_n th or $(n - a_n)$ th order statistics are independent and follows Normal and Gamma distribution respectively.

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